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New paleomagnetic investigations of the Emeishan basalts in NE Yunnan, southwestern China: Constraints on eruption history

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ABSTRACT

Paleomagnetic and magnetostratigraphic investigations have been conducted on two consecutive sections of the Emeishan trap at Shuimo (SM) and Dadi (DD) in northeastern Yunnan province, southwestern China. Rock magnetic analysis indicates that the magnetic carrier of the basalts is dominantly low-Ti magnetite. Stepwise thermal demagnetization isolates a stable characteristic remanent magnetization (ChRM) at high temperatures (400–585 °C). The ChRMs (301/323) at SM section are northeasterly upward of normal polarity, with a mean direction of $D_s/I_s = 11.0/-5.2°$ ($N = 12$, $k = 53.7$, $\alpha_{95} = 6.0°$). At DD section, the ChRMs are southwesterly downward of reversed polarity, with a mean direction of 205.9/10.0° ($N = 5$, $k = 78.4$, $\alpha_{95} = 8.7°$). The ChRMs from the basaltic gravels from the overlying conglomerate layer in late Permian Xuanwei Formation at DD section pass the conglomerate test, suggesting the primary nature of magnetization. Together with other magnetostratigraphic studies (e.g., N–R pattern) of the Emeishan basalts and the known radiometric ages (ca 260 Ma), it suggests that the eruption period of the Emeishan large igneous province is less than 3 Myr.

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1. Introduction

The Emeishan trap is one of the late Permian large igneous provinces (LIPs), covering an area of more than 2.5×10^6 km^2 (or $\sim 0.3 \times 10^6$ km^3) in the western margin of the South China Block (SCB) in the southwestern China (Xu et al., 2001; He et al., 2003; Fig. 1a). Previously, paleomagnetic studies were conducted on the Emeishan basalts to understand the tectonics and late Permian paleomagnetic pole of the SCB (McElhinny et al., 1981; Zhang, 1984; Lin et al., 1985; Zhou et al., 1986; Huang et al., 1986; Liu et al., 1987; Zhao and Coe, 1987; Van der Voo et al., 1993; Ma et al., 1993; Huang and Opdyke, 1995). It has been noted in early studies that both north–northeasterly shallow-up (normal) directions and southwesterly downward (reversed) directions, but the two directions failed reversal tests. This casted some suspicions on the primary magnetization recorded by Emeishan basalts. Several assumptions, e.g., local differential rotation and remagnetization, were proposed to explain the lack of anti-polarity (e.g., Huang, 1987; Zhou et al., 1986; Van der Voo et al., 1993; Wang and Van der Voo, 1993; Wang et al., 1993; Huang and Opdyke, 1998; Zheng et al., 2010). For example, Wang and Van der Voo (1993) and Van der Voo et al. (1993) suggested the normal polarity directions represent a younger remagnetization, whereas, Huang and Opdyke (1998) thought it is more likely due to the rotation between different localities.

The mantle plume origin was proposed in late 1990's to explain the formation of Emeishan LIP (e.g., Chung and Jahn, 1995), where short eruption duration is one of the critical bases (e.g., Xu, 2002). Magnetostratigraphic investigations was first attempted by Huang and Opdyke (1998) and followed by many others (e.g., Ali et al., 2002; Zheng et al., 2010). On the other hand, geochronological investigations gave rather controversial results. SHRIMP zircon U–Pb dating yielded a duration of 257–260 Ma (He et al., 2007; Zhou et al., 2002, 2008; Fan et al., 2008; Xu et al., 2008). Others suggested a longer duration of 10 Ma (251–260 Ma) on the basis of Ar^{40}–Ar^{39} and SHRIMP zircon U–Pb dating on three Emeishan basalts (Fan et al., 2004). Shellnutt et al. (2008) recently reported a much younger SHRIMP zircon U–Pb age (242 ± 2 Ma) for one Emeishan-basalt-like dyke in Panxi region.

In this paper we report new rock magnetic and magnetostratigraphic study on two consecutive sections of Emeishan LIP, Shuimo (SM) and Dadi (DD), in the northeastern Yunnan province, southwestern China. The aims of this study are to examine the reliability of magnetization and constrain the Emeishan basalt eruption period. Possible reasons of the lack of antipolarity have been discussed. Together with available geochronologic data, magnetostratigraphic correlations in the whole area suggest a short duration (<3 Myr) of the Emeishan basalt eruption.

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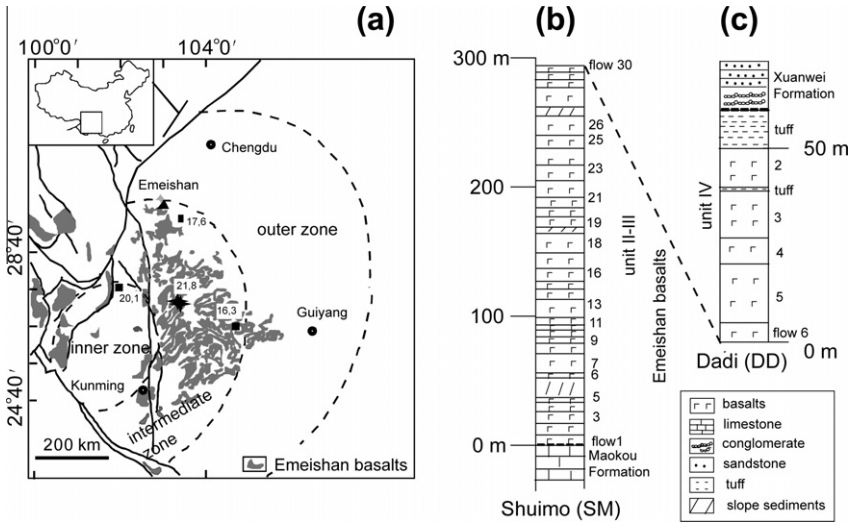


Fig. 1. (a) Simplified geological map of the distribution of the Emeishan basalts (after He et al., 2003). Dashed curves separate the inner, the intermediate and the outer zones of the large igneous province. The stars indicate the sample sections in this study and the squares indicate other magnetostratigraphic study localities (the numbers imply the references in Table 2). (b) and (c) are lithology of Shuimo and Dadi sections, respectively.

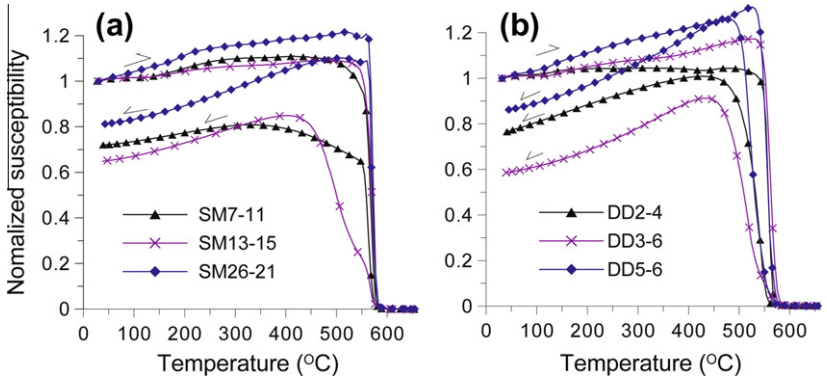


Fig. 2. Representative temperature dependent magnetic susceptibility (normalized by room susceptibility) curves on Shuimo (a) and Dadi (b) sections, respectively. The curves were obtained in argon atmosphere. Arrows indicate the heating/cooling processes.

2. Geological setting, sampling and methods

The Emeishan trap is divided into three zones, the inner, the intermediate and the outer, respectively (He et al., 2003; Fig. 1a). The Emeishan basalts Formation are unconformably underlain by the limestone of Lower Permian Maokou Formation, and are conformably overlain by the Upper Permian Xuanwei Formation. The Xuanwei Formation is composed of sandstones, mudstones and conglomerate with inter-bedded coal seams. A series of synclines with southeast to northwest striking axes is present in the Zhaotong area in northeastern Yunnan province. These synclines are part of the Mesozoic over-thrust fault system within the SCB (Yan et al., 2003), with Permian strata in the limbs and Triassic strata in the cores (Wang et al., 2006).

The Emeishan basalt Formation in the Zhaotong area is usually of more than 1000 m thick, divided into four stratigraphic units, I to IV, upwardly (Wang et al., 2006; Geological survey of China, 1978). Two basalt sections in the Zhaotong area in the intermediate zone were sampled in this study, one is near the town of Shuimo (SM; 27.28°N, 103.34°E) and the other is ~25 km west, near the village of Dadi (DD; 27.22°N, 103.54°E; Fig. 1a). The SM basalt sequence is composed of the units II and III of the Emeishan basalt Formation, and the DD is of the unit IV. Previous field survey showed that the DD sequence was consecutively emplaced on the SM sequence (Geological survey of China, 1978; Fig. 1).

The thickness of the SM basaltic sequence is ~300 m, containing at least 30 flows, with each flow ranging from 5 to 15 m (Fig. 1b). A lava flow usually starts with pyroclastics, followed by aphyric and/or (plagioclase-) porphyritic basalts and ending up with amygdaloidal and/or vescular basalts. Total 331 paleomagnetic samples were sampled from the SM sequence, and the bedding attitude (strike/dip = 209/10°) was determined on one representative pyroclastics. The DD basaltic sequence is ~50 m thick. Five lava flows can be recognized with the occurrence of layers of purple-reddish tuff in the upper part (Fig. 1c), and 81 paleomagnetic samples were drilled. A thick conglomerate bed of the Upper Permian Xuanwei Formation overlies the DD basaltic sequence, with sub-rounded to rounded, basaltic-dominated 10–50 cm cobbles cemented by fine-grained silty matrix. The bedding attitude of the overlying conglomerate bed (strike/dip = 357/20°) is used for the tilt correction of the DD section. Eleven cobbles were sampled for a conglomerate test. Both magnetic and sun compasses were used for in situ core orientation, and less than 5° difference was found between the two orientations. The samples were cut into 2-cm cylinder specimens in the laboratory and fresh off-cuts were for rock magnetic experiments.

To characterize the composition and domain state of the magnetic carriers, thermo-magnetic analysis of temperature-dependent susceptibility from room temperature to 650 °C is conducted in argon using a multifunction Kapa-bridge equipped

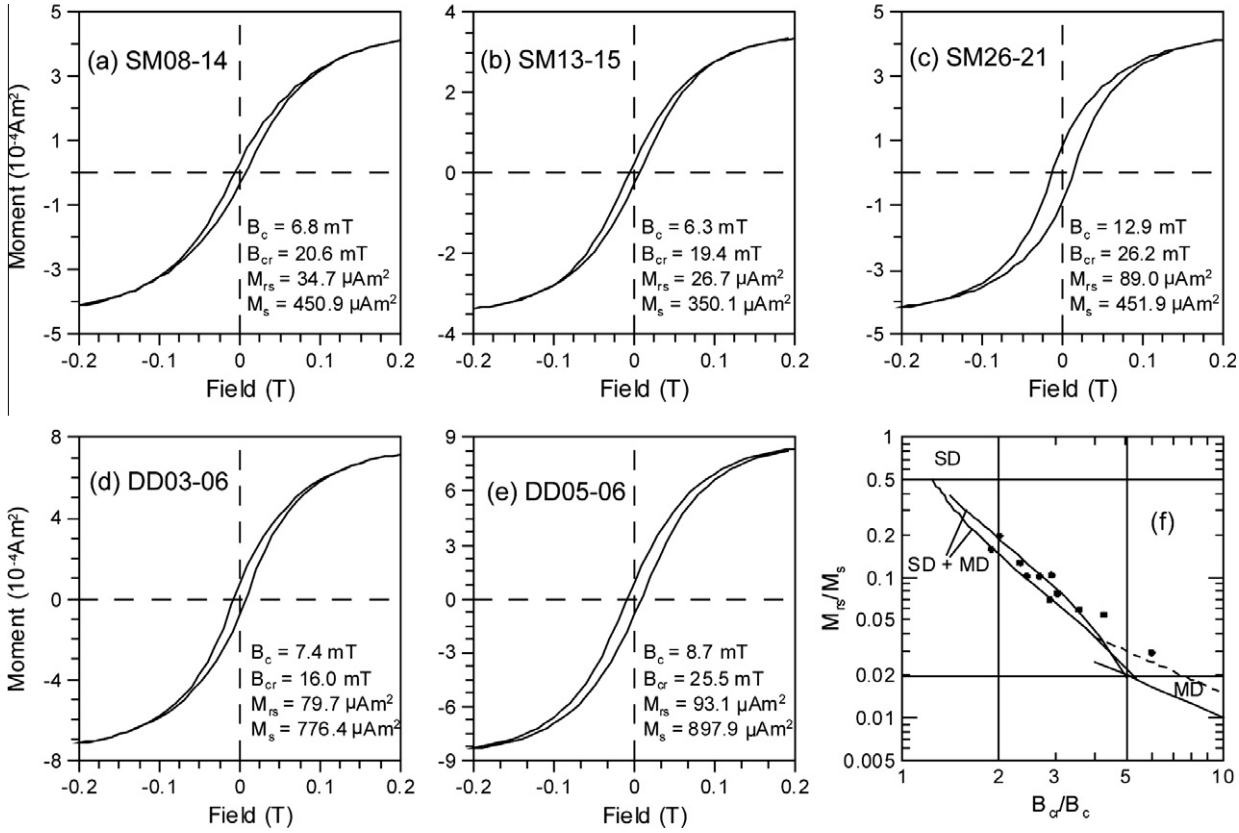


Fig. 3. (a–c) and (d, e) Representative hysteresis loops measured at room temperature after paramagnetic correction on Shuimo and Dadi sections, respectively. The saturation field is ±1 T. (f) The magnetic hysteresis parameters on the two sections are plotted on a Day diagram (Day et al., 1977; Dunlop, 2002). Abbreviations: B_{cr}, coercivity of remanence; B_c, coercivity; M_{rs}, saturate remanent magnetization; M_s, saturation magnetization; SD, single domain; MD, multidomain.

with a CS-3 high-temperature furnace (AGICO) in a field of 200 A/m with the frequency of 976 Hz. Hysteresis loops and acquisition of the remanence coercivity (B_{cr}) were measured on MicroMag 3900 Vibrating Sample Magnetometer (VSM) in room temperature with field ranges of ± 1T. Hysteresis parameters (M_{rs}/M_s versus B_{cr}/B_c) were plotted on a Day plot (Day et al., 1977; Dunlop, 2002).

All cylinder specimens were subjected to progressive thermal demagnetization in 13–15 steps up to 550 or 585 °C in an ASC TD-48 oven in the magnetically shielded room (residual field < 300 nT). The remanence was measured using a 2G-755R cryogenic magnetometer. To compare with the thermal demagnetization behavior, five specimens from different flows on the DD section were subjected to progressive alternating field (AF) demagnetization in 15 steps up to 70 mT. All the characteristic remanent magnetization (ChRM) directions were determined using principal component analysis (Kirschvink, 1980) or using PaleoMag software (Jones, 2002), within the mean angular deviations (MADs) less than 10°. Fisher statistics (Fisher, 1953) were employed to compute the site mean directions and Enkin's PMGSC software was used to calculate the angular deviation between the mean directions. The equal area projection of the directions is plotted with PmagPy programs (Tauxe, 2010).

3. Results

3.1. Rock magnetic results

The temperature-dependence of magnetic susceptibility curves show the susceptibility sharply drops at ~585 °C (Fig. 2), indicating that (low-Ti) magnetite is dominant magnetic carrier for the SM and DD sections. The susceptibility of the cooling curves is lower than that of the heating curves, which may be due to the irreversible oxidation of magnetite into hematite by heating. Room temperature hysteresis loops show that all the specimens from the two sections were saturated in less than 0.2 T (Fig. 3a–e), which confirms that (low-Ti) magnetite is the main magnetic contributor. The M_{rs}/M_s versus B_{cr}/B_c data (Fig. 3f) shows that the bulk magnetic properties generally follow the SD–MD trends (Day et al., 1977; Dunlop, 2002).

3.2. Paleomagnetic results

3.2.1. SM section

The NRM intensity of the basalt samples from the SM section varies from 0.03 to 13.6 A/m. The NRM directions are rather scattered (Fig. 4a). The stepwise thermal demagnetization revealed that a majority of specimens have two magnetic components: a low temperature component (LTC) and a characteristic magnetic remanent (ChRM) direction (Fig. 5). The LTC was generally removed below 400 °C and generally weighted a small amount in NRM (Fig. 5a, e, f). The LTC directions are scattered, however, falling close to the present magnetic field ($D/I = 359/41.5°$), with the mean direction of $D/I = 4.8/40.2°$ ($\alpha_{95} = 8.9°$) in the geographic coordinate (Fig. 4b). The ChRMs were well defined above 400 °C, unblocking up to 585 °C (Fig. 5). The ChRM directions (301 out of 323 specimens) are fisherian distributed, departing significantly from present magnetic field (Fig. 4c–f). All ChRMs are northeasterly with very shallow, mostly negative inclination in the stratigraphic coordinates (Fig. 4d and f).

F statistic is applied to compare the contiguous flow mean directions (Bultler, 1998). When two adjacent flow-mean directions are not distinct at 95% level significance, they are combined

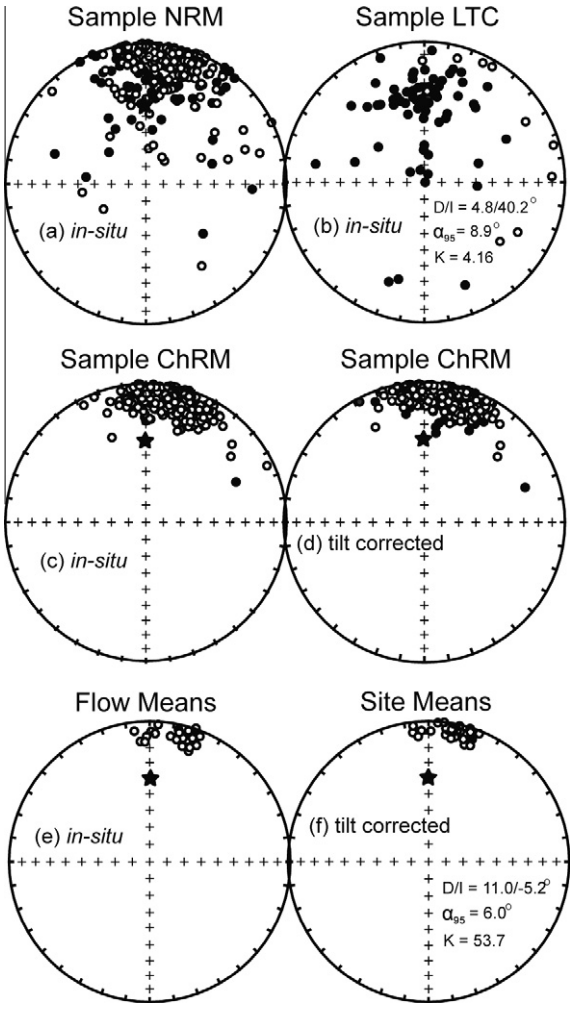


Fig. 4. Equal-area projections of sample NRM (a), low-temperature component (b), ChRM before (c) and after tilt correction (d), flow mean (e) and site mean (f) directions from the Emeishan basalts on Shuimo section. Open (solid) circles represent lower (upper) hemisphere projection. The star denote present geomagnetic magnetic field.

as one site. For SM section we obtain 12 paleomagnetically independent directions, with the mean direction of $D_s/I_s = 11.0/-5.2°$, $\alpha_{95} = 6.0°$ (Fig. 4f). Detailed statistic paleomagnetic results from SM section are given in Table 1.

3.2.2. DD section

The NRM intensity of the DD section samples varies from 0.41 to 12.0 A/m and their directions are rather scattered (Fig. 6a). The LTCs are also scattered, generally falling close to the present magnetic field (Fig. 6b).The scatter in NRMs is dramatically reduced after removing of the LTCs (Fig. 6c). The ChRM directions (58 out of 81 specimens) are all southwesterly with mostly downward in the stratigraphic coordinates (Fig. 6c), generating the mean tilt corrected directions $D_s/I_s = 205.9/10.0°$, $\alpha_{95} = 8.7°$ (Fig. 6d). The Zijderveld plots of the thermal demagnetization behavior (Fig. 6e–g) further reveals that LTCs were generally removed below 420 °C, and the ChRM directions were generally isolated at temperatures 420–585 °C. The AF demagnetization on representative samples also shows two magnetic components (Fig. 6h–j): a low coercivity component was generally removed below 10 mT and a ChRM direction was isolated between 15 and 70 mT. Detailed statistic paleomagnetic results of this section are given in Table 1.

Because of the shallow inclination, the declination is used to diagnose the polarity. Therefore, the SM samples with northeastward declinations recorded a normal polarity and on the other hand the DD samples with southwestward declinations recorded reversed polarity.

3.2.3. Paleosecular variation (PSV)

The angular dispersion of the virtual geomagnetic poles (VGPs), *S*, is commonly used as a parameter to estimate the influence of the paleomagnetic secular variation (PSV). *S* is defined as $S = \left(\frac{1}{N-1}\sum_{i=1}^{N}\Delta i^2\right)^{1/2}$, in which Δi is the angle between single (i) and the calculated mean direction. N is the number of sites, generally more than 5 (Bultler, 1998). At low latitude in the past 5 Ma, *S* is considered reasonable from 10 to 14 (e.g., McElhinny and McFadden, 1997; Biggin et al., 2008). The *S* value derived from SM section is 10.8 and that from the reversed DD directions is 9.2.

3.2.4. Reversal test

The anti-polarity was tested on the normal and reverse directions given in Table 1. The angular difference between two mean directions of SM and DD sections is 15.5°, more than the critical value, 10.3°, suggesting a negative McFadden and McElhinny's reversal test (McFadden and McElhinny, 1990). The lack of anti-polarity between the normal and reverse directions is commonly observed elsewhere in the Emeishan basalt province (e.g., Huang and Opdyke, 1998; Zheng et al., 2010).

3.2.5. Conglomerate test

Paleomagnetic results from eleven basaltic cobble samples of the overlying conglomerate of the Xuanwei Formation at the DD section for a conglomerate test were shown in Fig. 7. These cobbles were magnetized with the NRM intensity in the range from 0.005 to 4.18 A/m. The NRM was generally consisted of two magnetic components: a LTC dominated the NRM that was removed below 400 °C, and a ChRM was isolated at temperatures 400–585 °C (Fig. 7a–c). Both NRM and ChRM directions are scattered in geographic coordinate (Fig. 7d and e). The resultant vector *R* (6.23), of the eleven ChRM directions is slightly smaller than the 99% critical value, R_0 (6.25), for Watson's randomness test (Watson, 1956), which demonstrates that these directions pass the conglomerate test at the 99% confidence level. This indicates that the ChRM of the cobbles was acquired earlier than the formation of the conglomerate and without significant secondary remagnetization. This observation suggests that the magnetization in underlying Emeishan basalts is primary, in consistent with many other previous studies (e.g., Zheng et al., 2010; Ali et al., 2002; Huang and Opdyke, 1998; Ma et al., 1993).

4. Discussions

4.1. Nature of magnetization

As described above, most of the studied basaltic samples have two magnetic components. The LTC directions fall close to the present Earth's magnetic field in geographic coordinates and thus of the viscous magnetization origin. The high temperature component isolated from the individual conglomerates in the Upper Permian Xuanwei Formation at the upper DD section passes the conglomerate test, suggesting that the primary magnetization has been well preserved since the late Permian. This is strong evidence demonstrating that the isolated ChRMs from the DD section and the SM section are primary, which is consistent with numerous previous paleomagnetic investigations on the Emeishan basalts at other sections with positive fold and conglomerate tests

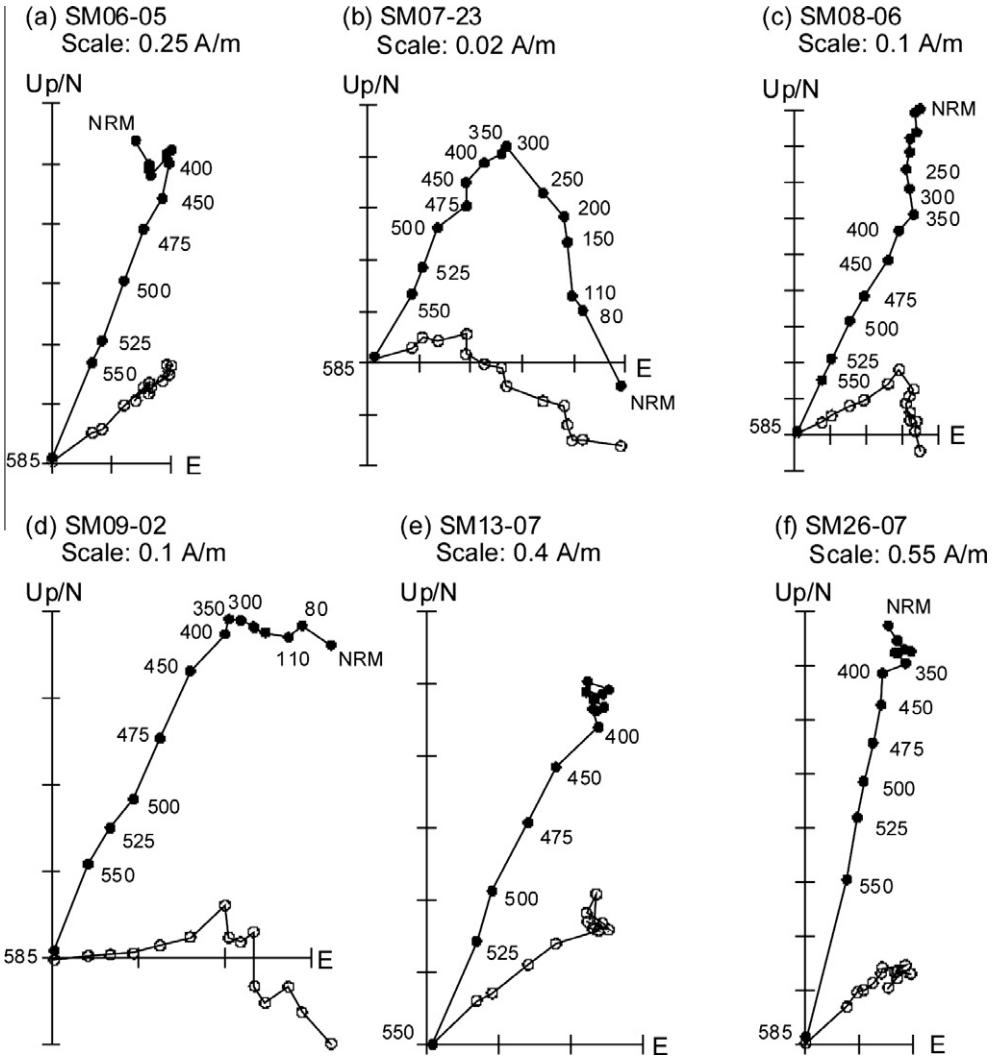


Fig. 5. Representative orthogonal plots (Zijderveld, 1967) of stepwise thermal demagnetization behavior in geographic coordinates on the Shuimo section. Open (solid) circles represent projections on to vertical (horizontal) plane. Treatment levels are in degrees Celsius.

Table 1
Statistic paleomagnetic results on site level on Shuimo and Dadi sections.

Site	n/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	R	K	α_{95} (°)	VGP_latt (°N)	VGP_long (°E)
Dadi (DD) section (27.22°N, 103.54°E), Strike/dip = 357/20°										
DD2	10/12	213.2	15.7	215.8	3.4	9.9	71.0	5.8	45.0	227.7
DD3	11/14	200.2	16.6	204.2	8.0	10.8	56.5	6.1	51.0	243.0
DD4	10/12	197.6	19.3	202.7	11.4	9.9	146.2	4.0	50.4	246.5
DD5	12/20	197.0	13.2	200.2	5.7	11.7	34.0	7.5	54.2	247.4
DD6	15/23	197.8	29.8	206.5	21.2	14.8	57.2	5.1	44.0	246.1
Section mean		**201.2**	**19.0**	**205.9**	**10.0**	**4.9**	**78.4**	**8.7**	**49.2**	**241.8**
Shuimo(SM) section (27.28°N, 103.34°E), Strike/dip = 209/10°										
SM-12	30/30	11.0	−11.2	9.4	−7.9	29.3	44.1	4.0	57.5	265.7
SM-11	11/13	21.9	−11.7	20.0	−10.3	10.7	30.8	8.4	52.2	249.6
SM-10	26/29	2.4	−5.9	1.9	−1.3	25.2	30.4	5.2	62.0	279.3
SM-9	14/16	−6.7	−7.1	352.7	−1.2	13.7	40.2	6.3	61.2	298.7
SM-8	49/51	−1.8	−12.4	356.7	−7.2	48.1	55.3	2.8	58.9	289.7
SM-7	14/15	12.9	−6.4	12.0	−3.5	13.7	42.2	6.2	58.8	259.7
SM-6	28/29	17.7	−15.2	15.3	−13.1	27.4	41.7	4.3	53.0	257.5
SM-5	20/22	16.9	−1.0	16.9	1.1	19.2	24.4	6.7	58.7	249.3
SM-4	67/74	17.3	−8.2	16.1	−6.1	65.1	34.9	3.0	55.9	253.7
SM-3	20/21	19.3	−17.6	16.4	−15.7	19.8	89.1	3.5	51.3	256.8
SM-2	10/11	12.3	4.2	13.3	7.0	9.8	58.8	6.4	63.0	252.9
SM-1	12/12	21.8	−4.4	21.2	−3.1	11.6	29.4	8.1	54.7	244.6
Section mean		**12.1**	**−8.2**	**11.0**	**−5.2**	**11.8**	**53.7**	**6.0**	**58.3**	**262.1**

N/n: numbers of the conducted/chosen samples. D_g (D_s), I_g (I_s): declination and inclination before (and after) tilt correction. R: reluctant radius. K: Fisher precision parameter of the mean directions. α_{95}: radius of the 95% confidence circle about the mean direction. VGP: abbreviation of virtual geomagnetic poles.

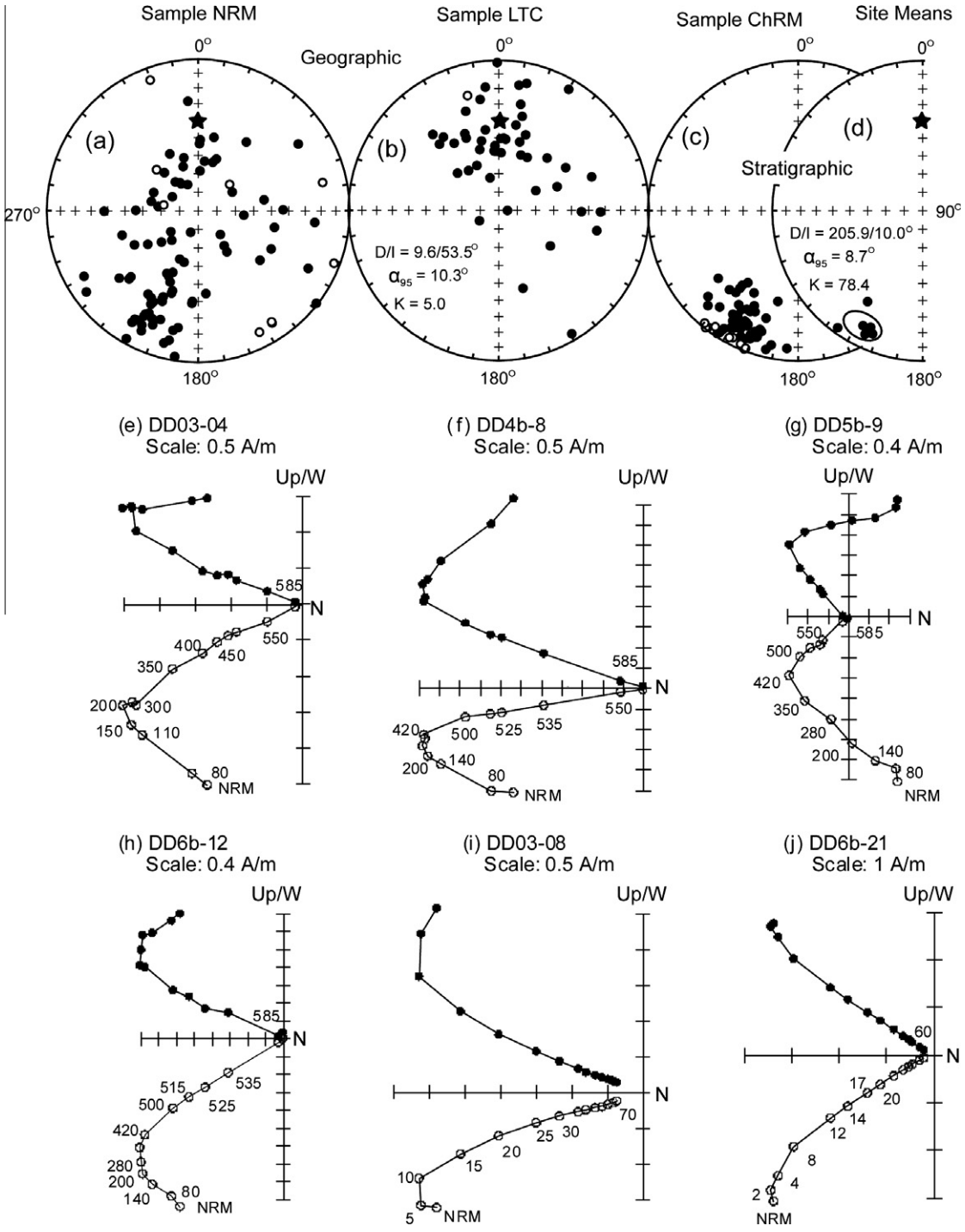


Fig. 6. Equal-area projections of sample NRM (a), low-temperature component (b), ChRM (c) and site mean directions (d) from the Emeishan basalts on Dadi section. Symbols as in Fig. 4. (e–g) and (h–j) Representative orthogonal plots of thermal and alternating field demagnetization behavior on the Dadi section, respectively. Coordinates and symbols as in Fig. 5, treatment levels in (e–g) are in degrees Celsius and (h–j) in millitesla.

(e.g., Huang and Opdyke, 1995, 1998; Ali et al., 2002; Zheng et al., 2010).

Some authors have argued that the normal polarity directions observed from the Emeishan basalts are later remagnetization, only the reverse directions are primary (Wang and Van der Voo, 1993; Van der Voo et al., 1993). This opinion based mainly on two scenarios. First, microcopy investigation in Van der Voo et al. (1993) reveals the normal polarity samples have secondary growth magnetite; second, paleomagnetic poles from the normal directions is closer to the Triassic pole of South China Block than that from the reversed directions. This study suggests the normal polarity recorded in the SM samples is primary magnetization. In addition to a positive conglomerate test on the overlying Xuanwei Formation, firstly, our microscopy observation did not show any hint of secondary magnetite on the SM samples. The rock magnetic experiments demonstrated the main magnetic carriers of both the SM and DD sections are (low-Ti) magnetite (Fig. 2). Secondly, all the ChRM was resolved in high temperatures after removing the

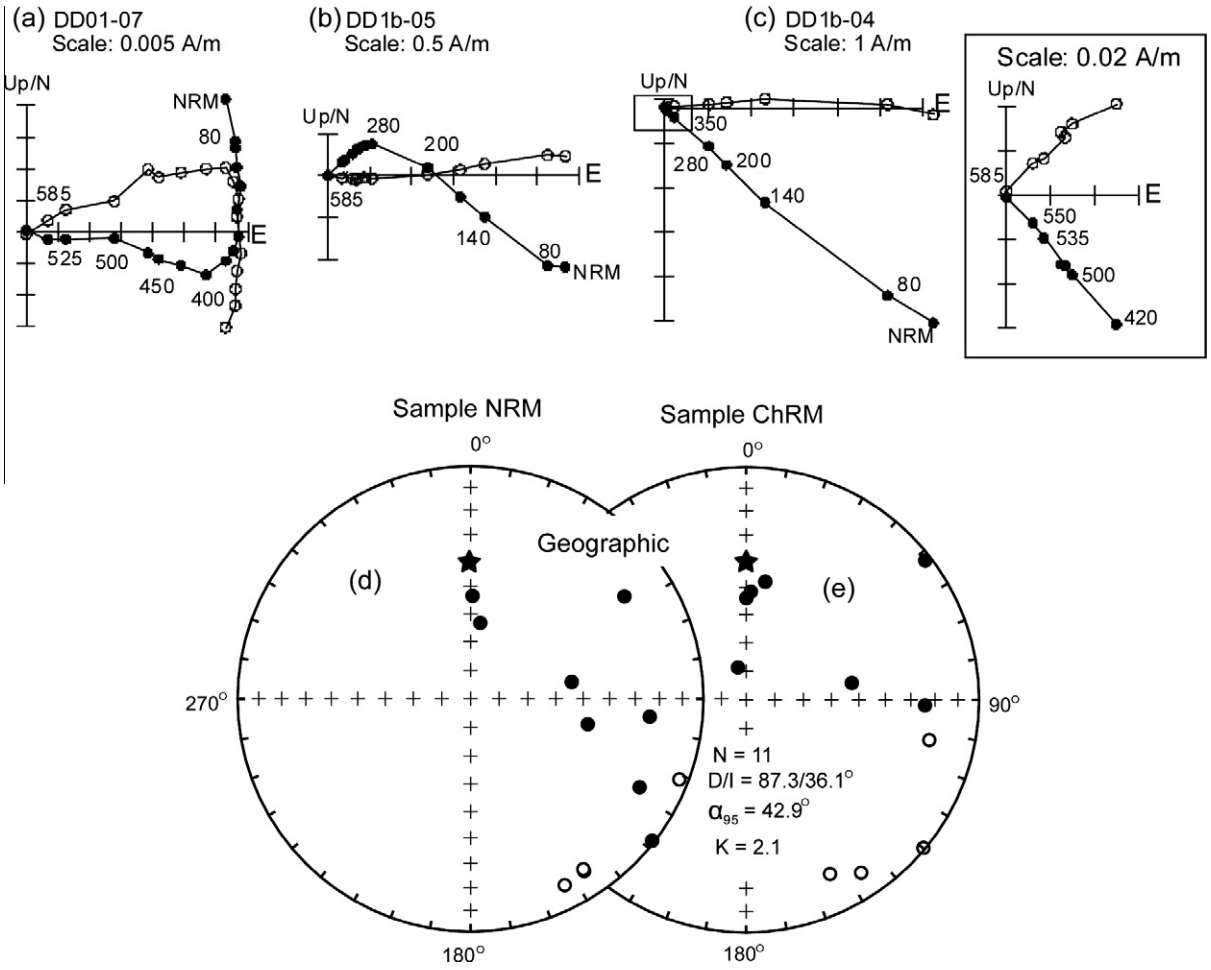


Fig. 7. (a–c) Representative orthogonal plots of thermal demagnetization results of the basaltic cobbles on the Dadi section. Square in (c) indicates the enlargement of the high temperature behavior. Coordinates and symbols as in Fig. 5. (d and e) Equal-area projections of sample NRM (d) and ChRM (e) in geographic coordinates of the cobbles on the Dadi section. Symbols as in Fig. 4.

Table 2
Paleomagnetic results from the Emeishan basalts.

No.	Locality	Lat. (°N)	Long. (°E)	*N*	*n*	D_s (°)	I_s (°)	*K*	α_{95} (°)	Polarity	Reference
1	Luomapu	27.67	101.88	19	168	250.1	23.7	27.2	6.5	R	Zheng et al. (2010)
2	Gongmushan	27.37	101.54	12	101	279.2	14.9	74.2	5.1	R	Zheng et al. (2010)
3	Duge	26.4	104.7	3	18	232.3	21.6	42.7	19.1	R	Huang and Opdyke (1998)
4	Huidong	26.7	102.9	2	6	199.9[a]	3.5[a]	17.0[a]	24.0[a]	R	Huang et al. (1986)
5	Bingchuan	25.6	100.6	3	9	242.0	1.3	25.9	15.9	R	Huang et al. (1986)
6	Shuangqiao	29.34	103.29	1	4	164.0	−17.0	32.2	16.4	R	Ali et al. (2002)
7	Ertan	26.8	101.8	?	12	234.5	22.9	24.4	9.0	R	Zhou et al. (1986)
8	Dadi	27.22	103.54	5	57	205.9	10.0	78.4	8.7	R	This study
9	Emei	29.6	103.4	3	24	16.4	−4.5	71.2	3.5	N	Ma et al. (1993)
10	Emei	29.7	103.5	?	46	38.4	−11.8	53.0	4.8	N	Liu et al. (1987)
11	Zhaojue	28.1	102.9	?	19	18.1	−4.9	24.0	6.6	N	Liu et al. (1987)
12	Huidong	26.7	102.9	5	24	24.2[a]	−5.0[a]	40.6[a]	9.8[a]	N	Huang et al. (1986)
13	Meigu	28.3	103.0	?	10	32.5	−2.5	19.9	8.4	N	Zhou et al. (1986)
14	Kunming	25.0	102.7	10	57	25.5	−11.9	63.5	6.1	N	Van der Voo et al. (1993)
15	Weining	26.7	104.0	1	9	359.3	−6.0	27.6	10.0	N	Huang and Opdyke (1995)
16	Duge	26.4	104.7	19	122	39.8	0.6	133.5	2.9	N	Huang and Opdyke (1998)
17	Xinchung	29.27	103.30	7	39	16.7	8.9	23.8	14.0	N	Ali et al. (2002)
18	Ebian	29.27	103.27	6	37	14.5	10.3	131.7	5.9	N	Ali et al. (2002)
19	Qiaojia	27.07	102.98	9	116	357.9	−10.6	40.0	7.4	N	Liu and Zhu (2009)
20	Luomapu	27.7	101.9	8	82	44.0	−18.1	26.7	10.9	N	Zheng et al. (2010)
21	Shuimo	27.28	103.34	12	301	11.0	−5.2	53.7	6.0	N	This study

Lat., Long.: latitude and longitude of the sample locality. *N* (*n*): numbers of the sites (samples).D_s, I_s, mean declination and inclination in stratigraphic coordinates. *K*: Fisher precision parameter of the mean directions. α_{95}: radius of the 95% confidence circle about the mean directions.

[a] Recalculated from original data.

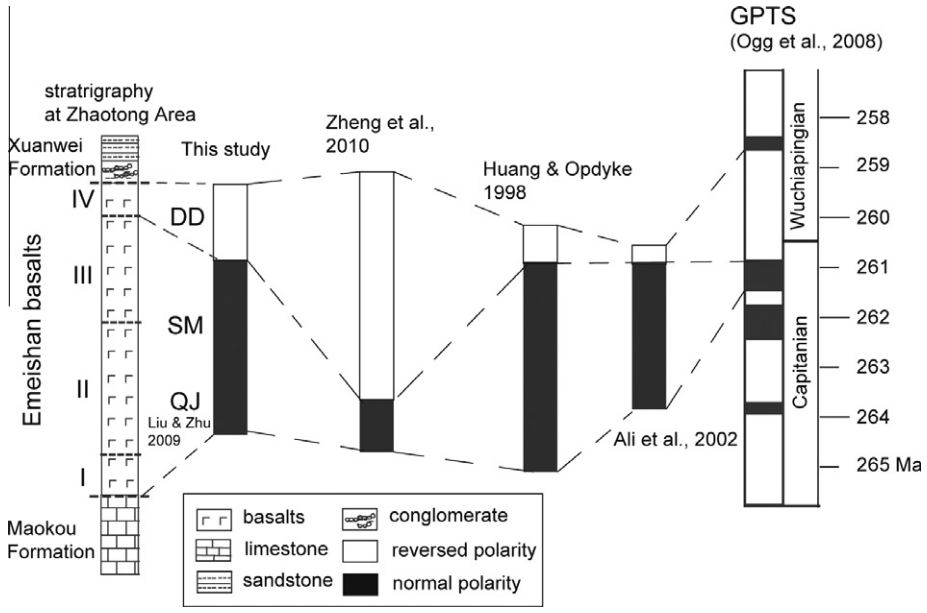


Fig. 8. Correlation of stratigraphy and magnetic sequences obtained in Zhaotong area from Qiaojia (QJ; Liu and Zhu, 2009), Shuimo (SM) and Dadi (DD) sections, and other magnetostratigraphic studies from Huang and Opdyke (1998), Ali et al. (2002) and Zheng et al. (2010), and to the Global polarity time scales (GPTs) (Ogg et al., 2008).

LTC component. The trajectories obtained from thermal demagnetization are perfectly linear and decay straight to the origin, suggesting that both the normal and reversed ChRM directions are unlikely to be contaminated by secondary magnetization. Thirdly, the pole position derived from the ChRMs of the basalt samples at SM section is different from any post-Permian paleomagnetic poles reported for SCB. Finally, twelve paleomagnetically independent mean normal directions from the SM section are unlikely induced from a late remagnetization event. Therefore, the isolated ChRMs from both SM and DD sections in this study are primary magnetization.

4.2. Tectonic rotation influence

The failure of the reversal test could arise from several reasons, e.g., errors in sample orientation, incomplete demagnetization, later remagnetization, inadequate sampling of PSV during either (or both) of the polarity intervals, the non-dipole magnetic fields or asymmetric reversals nature in the Late Permian, a regional differential tectonic rotation and the movement of the whole block (e.g., Bultler, 1998; Van der Voo et al., 1993). A systematic error in the sample orientation, incomplete demagnetization and remagnetization can be easily ruled out in this study. The normal and reversed polarity site-mean VGPs distributed within reasonable scatter ($S = 10.8$ and $S = 9.2$, respectively), suggesting the PSV is nearly averaged out. The bootstrap reversal test (Tauxe, 2010) shows the confidence intervals of the normal and reversed directions overlap for Z component but not for the X and Y component (data not shown), indicating the failure is mainly caused by the declination differences. There is no apparent evidence that this Late Permian field reversal is asymmetric. Thus, it leaves the local tectonic rotation or movement of the whole SCB during the N–R interval to be examined.

Enkin et al. (1992) suggested the SCB rotated ~20° counterclockwise during a short period near the Permian–Triassic boundary. Yang and Besse (2001) also suggested the SCB as a whole counterclockwise rotated in that period. We compiled paleomagnetic directions of Late Permian age of SCB yielded from the Emeishan basalts (Table 2). The large scatter of those paleomagnetic directions shows that the local tectonic rotation influences cannot be excluded. For instance, those data could be influenced by the south–north fault systems in the northern Yunnan.

To test the notion of a rotation during the N–R interval, we further analyzed exclusively the N–R directional data generated from the same/adjacent sections, where the potential local tectonic disturbance is minimized. Three sections, Duge (Huang and Opdyke, 1998), Luomapu (Zheng et al., 2010) and SM–DD (this study), which recorded both the normal and reversed polarities, are selected. There is an apparent declination deviation between the normal and reversed directions in all three sections. The declination differences from revered to normal directions are 11.4 ± 10.2°, 25.3 ± 9.1° and 14.2 ± 7.2° on the Duge, Luomapu and SM–DD sections, respectively. These limited results suggest that the sampling sites, even the whole Emeishan trap, may have had a counterclockwise rotation during the N–R reversal. However, more paleomagentic studies are needed to test this assumption.

4.3. Duration of the Emeishan basalt formation

Available zircon U–Pb ages on mafic rocks in the Emeishan LIP range from 257 to 263 Ma, peaked at ~260 Ma, which is considered to represent the main phase of the Emeishan flood volcanism (e.g., He et al., 2007; Zhou et al., 2002, 2008; Fan et al., 2008; Xu et al., 2008). The main phase of the Emeishan basalt formation is also temporally close to the end-Guadalupian mass extinction (i.e. 258 Ma or 260.4 Ma, Yin et al., 1992; Retallack et al., 2006), which led to the hypothesis that the Emeishan basaltic eruption is the main cause of the end-Guadalupian mass extinction (e.g., Zhou et al., 2002; Courtillot and Renne, 2003; Retallack et al., 2006).

Magnetostratigraphic works provide another approach to constrain the duration of the Emeishan volcanism. Our previous work on Qiaojia (QJ) section (~30 km south of the SM section) revealed the exclusively normal polarity on the unit I of Emeishan basalt (Liu and Zhu, 2009). This study has revealed that the units II–III of the Emeishan basalt on SM section of normal polarity and the unit IV on the DD section of reversed polarity (Fig. 8).Thus, the N–R polarity pattern determined from the QJ–SM–DD sections is likely representative to a whole Emeishan eruption period.

Previous magnetostratigraphic works at the Duge, Shuangqiao, Luomapu sections by Huang and Opdyke (1998), Ali et al. (2002)

and Zheng et al. (2010) reported a similar normal to reversed magnetic polarity sequence. Taking into account together the dominant ratiometric age of the Emeishan basalts (~260 Ma) and the geomagnetic polarity time scales (GPTS, Ogg et al., 2008), we can correlate the N–R polarity sequence to the upper Wuchiapingian to the lower Capitanian stage, which suggests a short duration less than 3 Myr in the whole Emeishan LIP (Fig. 8). The huge volume of the initial Emeishan basalts, e.g., 3.8×10^6 km^3 (Xu and He, 2007) and the short duration (<3 Myr) of the main basaltic activity provided the possibility that the Emeishan basaltic activity may cause paleoenvironment dramatically changed in a short time that may led to the end-Guadalupian mass extinction. On the other hand, the short eruption duration of the Emeishan basalt formation supports it as a mantle plume origin (e.g., Chung and Jahn, 1995; Xu, 2002).

5. Conclusions

Paleomagnetic investigation on the units II–III and unit IV of the Emeishan basalts at the SM section and DD section in NE Yunnan, generated reliable north–northeastern upward (normal) and southwesterly downward (reversed) ChRM directions, respectively. The established N–R magnetic polarity pattern from the QJ–SM–DD sequence is well consistent with previous magnetostratigraphic results of other sections and is thus representative to the whole Emeishan basaltic profile in the intermediate zone. By comparing the N–R polarity column with the GPTS (Ogg et al., 2008), it strongly suggests that the Emeishan trap was likely erupted within a short time (<3 Myr). Further direct radiometric age determinations on the Emeishan basalts and magnetostratigraphic investigations on the overlaying and underlying strata of the Emeishan basalts are still needed.

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